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AI-Enabled Early Autism Screening, Diagnosis Support & Post-Diagnosis Care Management Platform

Theme

Neurodevelopmental Care Innovation, Early Intervention, and Longitudinal Outcome Improvement

Global Autism Care Context

- Autism Spectrum Disorder (ASD) is a neurodevelopmental condition affecting social communication, behavior motor coordination, and sensory processing.
- According to the World Health Organization, 1 in 100 children globally is estimated to be on the autism spectrum with prevalence rising year over year.
- In high-income countries, diagnosis often occurs at 3–5 years of age; in low- and middle-income countries, diagnosis may be delayed beyond 6–8 years or missed entirely.
- Early intervention (before age 5) is clinically proven to improve speech, cognition, adaptive behavior, and independence, yet access to specialists is highly unequal.

Key takeaway: Autism outcomes are not limited by treatment efficacy but by delayed diagnosis, fragmented care, and lack of continuous monitoring.

Stakeholders Supported

- Pediatric Neurologists & Developmental Pediatricians
- Care Givers / Therapists (Speech, Occupational, Physiotherapy)
- Parents & Family Members
- Pediatric Patients (Infants to Adolescents)

Business Process in Scope

End-to-End Autism Care Continuum: Early Screening → Diagnostic Support → Therapy Planning → Progress Tracking → Outcome Optimization

Problem (Expanded – Real-World Context)

Autism diagnosis today is largely behavioral and observational, relying on tools such as ADOS (Autism Diagnostic Observation Schedule), caregiver questionnaires & clinician judgment. These methods, while clinically validated, are:

- Time-intensive (multiple sessions over months)
- Subjective (dependent on evaluator experience)
- Specialist-constrained (neurologists and therapists are scarce globally)

As a result:

- Many children miss the critical neuroplasticity window for early intervention.
- Families experience diagnostic uncertainty, stress, and fragmented care journeys.
- Post-diagnosis therapy plans are often generic, siloed, and poorly tracked.

Despite advances in AI, neuro-sensing, and behavioral analytics, autism care remains disconnected across diagnosis and long-term management.

Key Gaps Across the Autism Care Journey:

1. Delayed and Inconsistent Early Screening

What Happens Today

- Screening depends on parental concern and pediatric referral.
- Subtle early signs (reduced eye contact, atypical gestures) are often missed.

Why This Is a Problem

- Early behavioral markers appear as early as 6–12 months.
- Delayed screening leads to delayed therapy and poorer outcomes.

2. Subjective Diagnostic Assessments

What Happens Today

- Diagnosis relies heavily on clinician observation and checklists.
- Access to ADOS-trained specialists is limited.

Why This Is a Problem

- Inter-rater variability exists.
- Long wait times delay formal diagnosis.

3. Under-Utilization of Objective Behavioral & Neural Signals

What Is Possible but Rarely Used

- Eye-tracking patterns
- Gesture and motor coordination analysis
- EEG-based neural response signatures
- Facial micro-expression and attention metrics

Why This Is a Problem

- These signals provide objective, quantifiable biomarkers, yet are not integrated into routine care.

4. Fragmented Post-Diagnosis Therapy Management

What Happens Today

- Speech therapy, physiotherapy, and learning interventions are delivered in silos.
- Progress tracking is manual and inconsistent.

Why This Is a Problem

- Therapy plans are not dynamically adapted.
- Parents lack visibility into developmental progress and gaps.

5. Limited Continuous Monitoring & Outcome Measurement

What Happens Today

- Assessments are periodic and clinic-based.
- Day-to-day behaviors at home or school are not captured.

Why This Is a Problem

- Regression or stagnation is detected late.
- Therapy effectiveness is hard to measure objectively.

Systemic Impact Summary (Across regions):

- Autism care is **reactive instead of proactive.**
- Diagnosis is **late, subjective, and resource-constrained.**
- Therapy management is **disconnected and static.**
- Families carry the coordination burden.

This creates a strong global unmet need for an integrated, AI-enabled autism care platform. Lead-Up to an Automated AI-Driven Platform- A next-generation solution must support both diagnosis and long-term care, not just one stage.

How AI & Advanced Sensing Can Help

AI-Driven Screening & Diagnostic Support

- Computer vision to analyze:
  - Eye gaze, joint attention, facial expressions
  - Gestures, posture, and motor coordination
- EEG signal analysis to identify neural response patterns.
- Multimodal AI models combining behavioral + neural signals.
- Decision support tools for neurologists—not black-box diagnosis.

Post-Diagnosis Therapy & Progress Management

- Personalized therapy plans for:
  - Speech and language development
  - Physiotherapy and motor skills
  - Learning disability support
  - Social interaction training
- Continuous monitoring via caregiver and parent inputs.
- AI-driven progress tracking and gap identification.

Agentic AI Orchestration

- Screening Agent → flags early risk signals
- Clinical Support Agent → assists neurologists with evidence summaries
- Therapy Planning Agent → personalizes intervention plans
- Progress Monitoring Agent → tracks developmental gains and alerts regressions

Evaluation Criteria:

- Ability to integrate multimodal data (video, EEG, behavioral inputs).
- Accuracy and explainability of screening insights.
- Support for clinician decision-making (not replacement).
- Personalization and adaptability of therapy plans.
- Longitudinal progress tracking and gap reduction metrics.
- Ethical handling of pediatric data and parental consent.

Potential Approaches

- Use computer vision for gaze and gesture analysis.
- Apply signal processing and ML on EEG data (synthetic or simulated).
- Build multimodal fusion models for screening confidence scoring.
- Design agent-based workflows for diagnosis support and therapy management.
- Create caregiver and parent dashboards for progress visibility.

References & Study Material

Global Autism & Neurodevelopment Research

- **World Health Organization — Autism prevalence and care gaps**
  https://www.who.int/news-room/fact-sheets/detail/autism-spectrum-disorders
- **Centers for Disease Control and Prevention — Autism surveillance (ADDM Network)**
  https://www.cdc.gov/autism

AI & Autism Research

- Eye-tracking and gaze studies in early autism detection (Nature, JAMA Pediatrics)
- EEG biomarkers in ASD research (Frontiers in Neuroscience)
- Computer vision for behavioral analysis (IEEE, CVPR healthcare tracks)

Therapy & Intervention Evidence

- Early Start Denver Model (ESDM) studies
- Speech and occupational therapy outcome research
- Neuroplasticity and early intervention literature